



NORTHERN UNIVERSITY OF BUSINESS AND TECHNOLOGY

AI Study and Task Planner

UI/UX Design and Development Planning

Week 05 Submission - CSE4204 Mobile Computing Lab

Course	CSE4204 Mobile Computing Lab
Section	8B
Team	T03
Official Team	CSE4204-8B-T03
Project Type	Web-based AI academic productivity system

DOCUMENT GUIDE

Submission Overview

This document combines all mandatory Week 05 deliverables into one submission-ready report. High-fidelity UI designs created in Figma Make are used instead of separate low-fidelity wireframes, as permitted by the assignment brief.

UI/UX Design Complete	User Flow Complete
Development Roadmap Complete	Task Distribution Complete
Technology Setup Plan Complete	Figma Link Included
GitHub Link Included	

Document Sections

- Team and Project Information
- UI/UX Design Principles
- Design System and Major Screens
- Responsive Design Considerations
- User Flow Design
- Development Roadmap
- Team Task Distribution
- Final Technology and Setup Plan
- Repository and Design Links

PROJECT INFORMATION

Team and Project Information

Role	Name	Student ID
Team Leader	SAIM HASAN NAHID	11220320897
Backend Developer	AISHA SIDDIKA OISHEE	11220320893
Frontend Developer	EITY	11220320896
AI Integration Lead	MD KHALED HASAN RIHAD	11220320891

Project Summary

AI Study and Task Planner is a web-based system that helps university students manage courses, academic tasks, deadlines, priorities, and progress. It also provides AI-assisted study planning, task prioritisation, task breakdown, chat support, and saved study plans.

Primary Users and Scope

- Primary user: university student.
- Main scope: course management, task management, progress tracking, and AI assistance.
- Admin panel: not included because it is not part of the main implementation scope.
- Design output: high-fidelity, responsive, implementation-ready UI blueprint.

DESIGN RATIONALE

UI/UX Design Approach

1. Simple	Clear page hierarchy, familiar controls, and limited visual noise.
2. Consistent	Shared colours, cards, buttons, spacing, forms, and navigation.
3. Responsive	Desktop-first layouts with mobile-friendly forms and navigation.
4. Accessible	Readable contrast, descriptive labels, visible status states, and large tap targets.
5. User-focused	Academic tasks, deadlines, progress, and AI support remain central.

Requirement Coverage

Required Deliverable	Coverage
Major UI screens	Landing, authentication, dashboard, courses, tasks, progress, profile, and AI pages
User flow	Clear navigation flow included
Development roadmap	Weeks 06 to 11 included
Team task distribution	All four members assigned
Technology setup plan	Final stack and deployment plan included

UI DESIGN

Design System

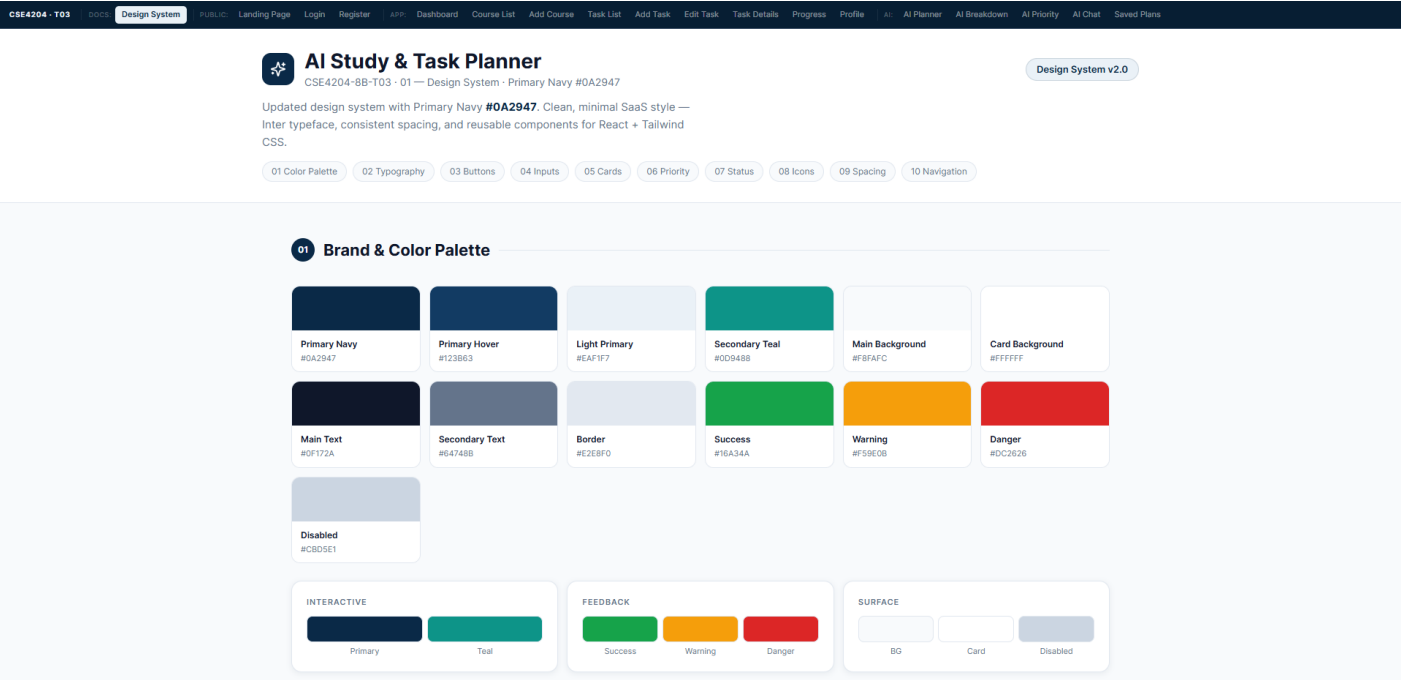


Figure 1. Brand colours, typography, components, states, spacing, and navigation system.

Landing / Home Page

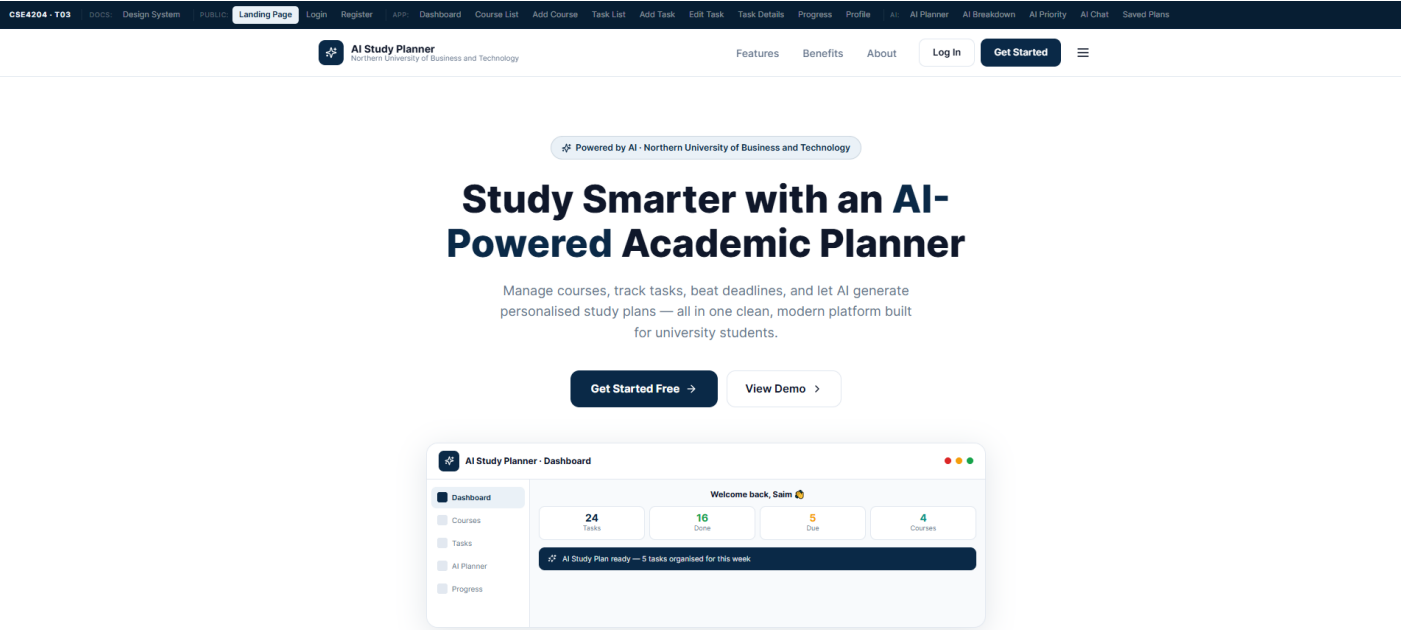


Figure 2. Public landing page with value proposition, feature navigation, and primary calls to action.

PUBLIC SCREENS

Authentication Screens

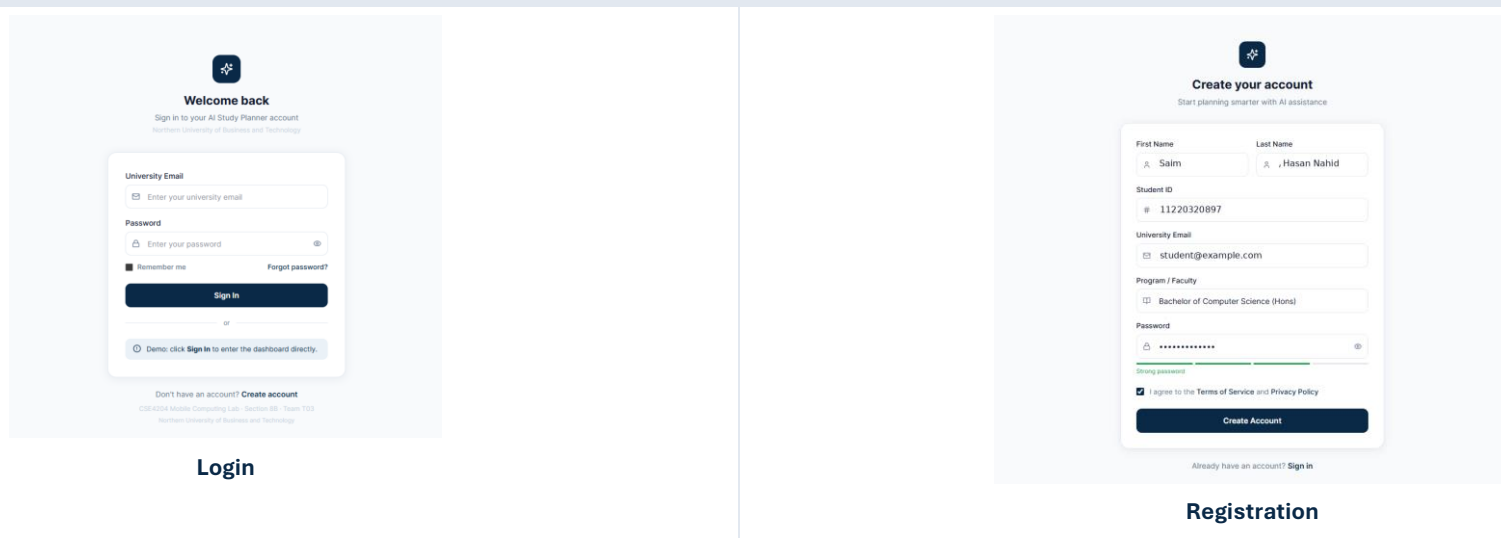
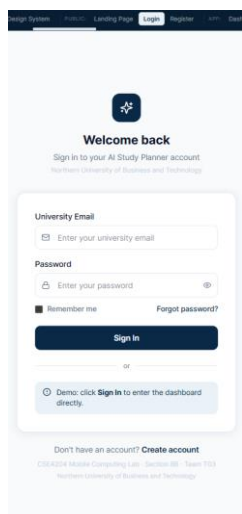


Figure 3. Authentication screens for secure account access and new-user registration.

RESPONSIVE UI

Responsive Design Example



Mobile Design Considerations

- Full-width form controls and large touch targets.
- Single-column layout for small screens.
- Readable typography with sufficient spacing.
- Mobile navigation can use a drawer or bottom navigation.
- Tables convert to stacked cards or horizontal scrolling.
- Buttons and actions remain visible without text overflow.

Figure 4. Responsive login layout demonstrating the mobile design approach.

UI DESIGN

Student Dashboard

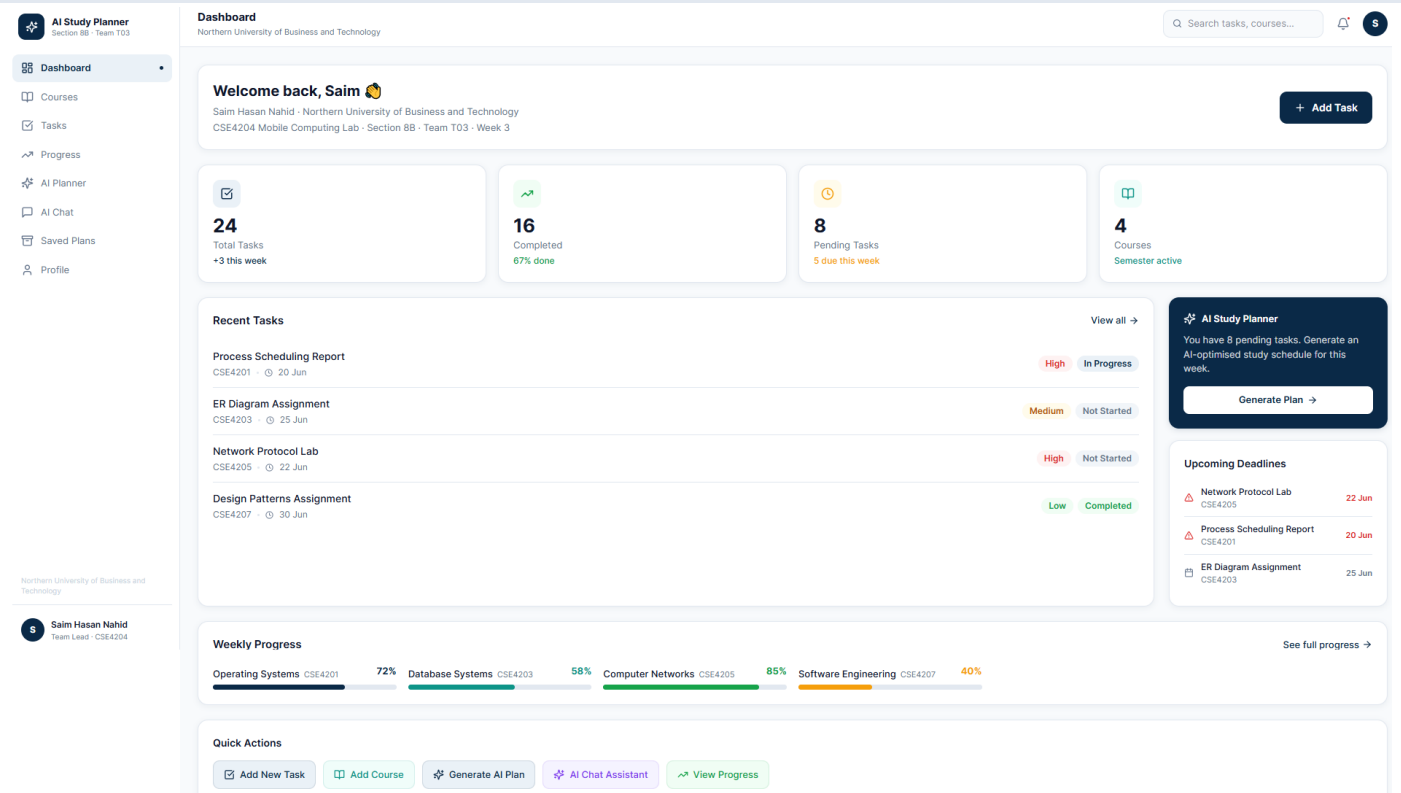


Figure 5. Dashboard summarising courses, tasks, deadlines, progress, and AI planning actions.

Course Management

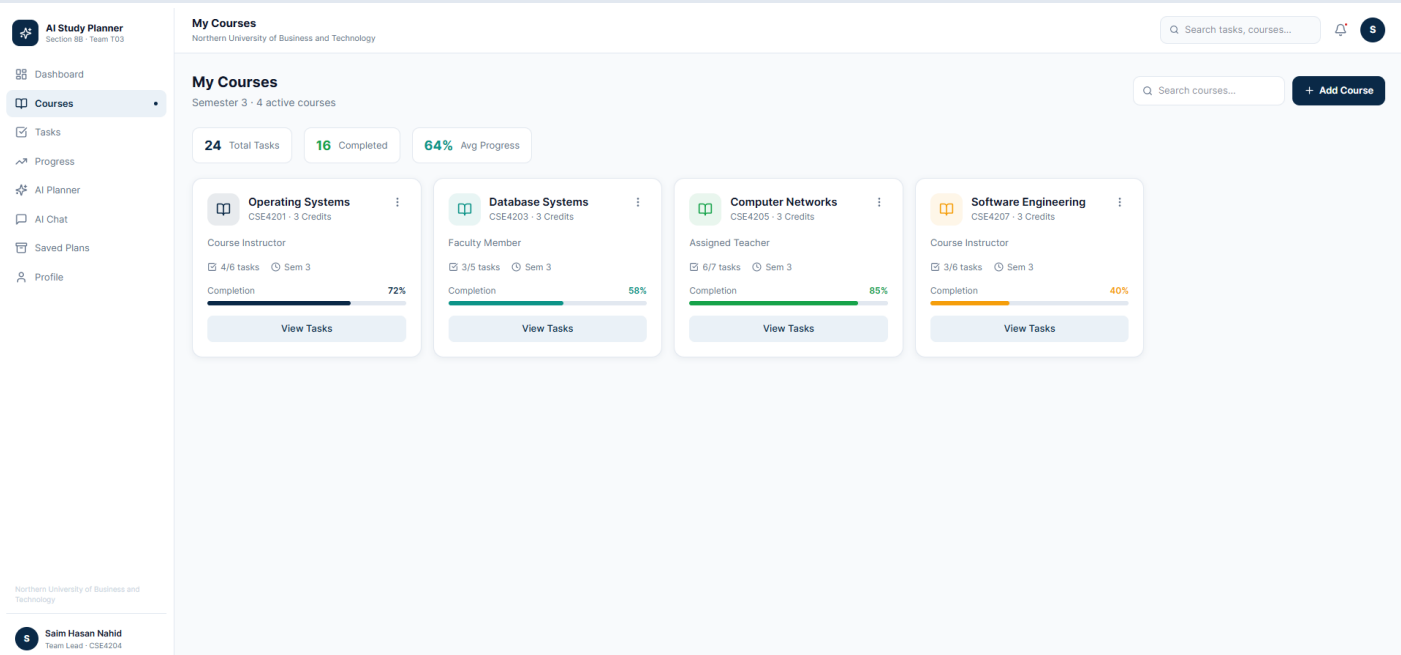


Figure 6. Course cards with task counts, completion progress, and quick navigation.

UI DESIGN

Add Course Form

The screenshot shows the 'Add Course' form in the AI Study Planner application. The form is titled 'Add New Course' and includes a sub-header 'Register a course to start tracking tasks and progress'. The form fields are as follows:

- Course Name ***: A text input field with a placeholder 'e.g. Operating Systems'.
- Course Code ***: A text input field with a placeholder '# e.g. CSE4201'.
- Credit Hours**: A text input field with a placeholder '3 Credits'.
- Instructor Name**: A text input field with a placeholder 'Enter instructor name'.
- Semester**: A dropdown menu with 'Semester 3 (Current)' selected.
- Colour Label**: A set of five colored circles (black, green, blue, yellow, red).
- Description (optional)**: A text area with a placeholder 'Brief course description or notes...'.

At the bottom of the form are two buttons: 'Cancel' and '+ Add Course'.

The sidebar on the left shows the navigation menu with 'Courses' selected. The top right corner has a search bar and a user profile icon.

Figure 7. User-friendly course creation form with validation-ready fields.

Task Management

The screenshot shows the 'My Tasks' list in the AI Study Planner application. The list is titled 'My Tasks' and includes a sub-header '5 tasks - 1 completed'. The list is filtered by 'All' tasks and sorted by 'High' priority. The tasks are as follows:

- Process Scheduling Report**: Status: High, In Progress. Due: 20 Jun 2025.
- ER Diagram Assignment**: Status: Medium, Not Started. Due: 25 Jun 2025.
- Network Protocol Lab Report**: Status: High, Not Started. Due: 22 Jun 2025.
- Design Patterns Implementation**: Status: Low, Completed. Due: 30 Jun 2025.
- Deadlock Detection Algorithm**: Status: Medium, Not Started. Due: 28 Jun 2025.

Each task entry includes a radio button, a title, a course name, a due date, a status indicator, and a set of action buttons (edit, delete, and a plus icon).

The sidebar on the left shows the navigation menu with 'Tasks' selected. The top right corner has a search bar and a user profile icon.

Figure 8. Searchable and filterable task list with status, priority, and CRUD actions.

UI DESIGN

Add Task Form

[← Back to Tasks](#)

Add New Task

Create a new academic task

Task Title *

e.g. Complete Assignment 3 – Process Scheduling

Course *

Select course

Deadline *

06/20/2025

Priority

Medium

Status

Not Started

Description / Notes

Add task description, requirements, or notes...

Cancel

+ Add Task

Figure 9. Task creation form for title, course, deadline, priority, status, and notes.

UI DESIGN

Task Details and Actions

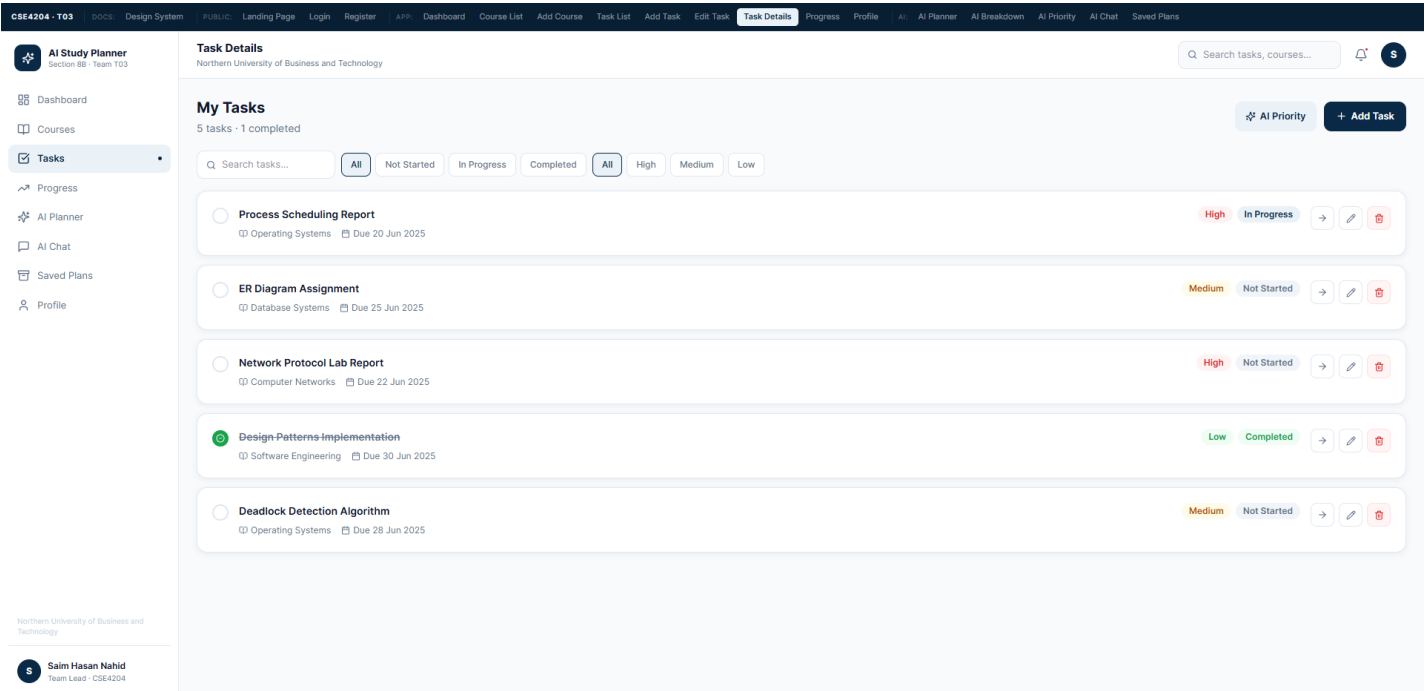


Figure 10. Task details and list actions, including view, edit, complete, and delete controls.

Progress Overview

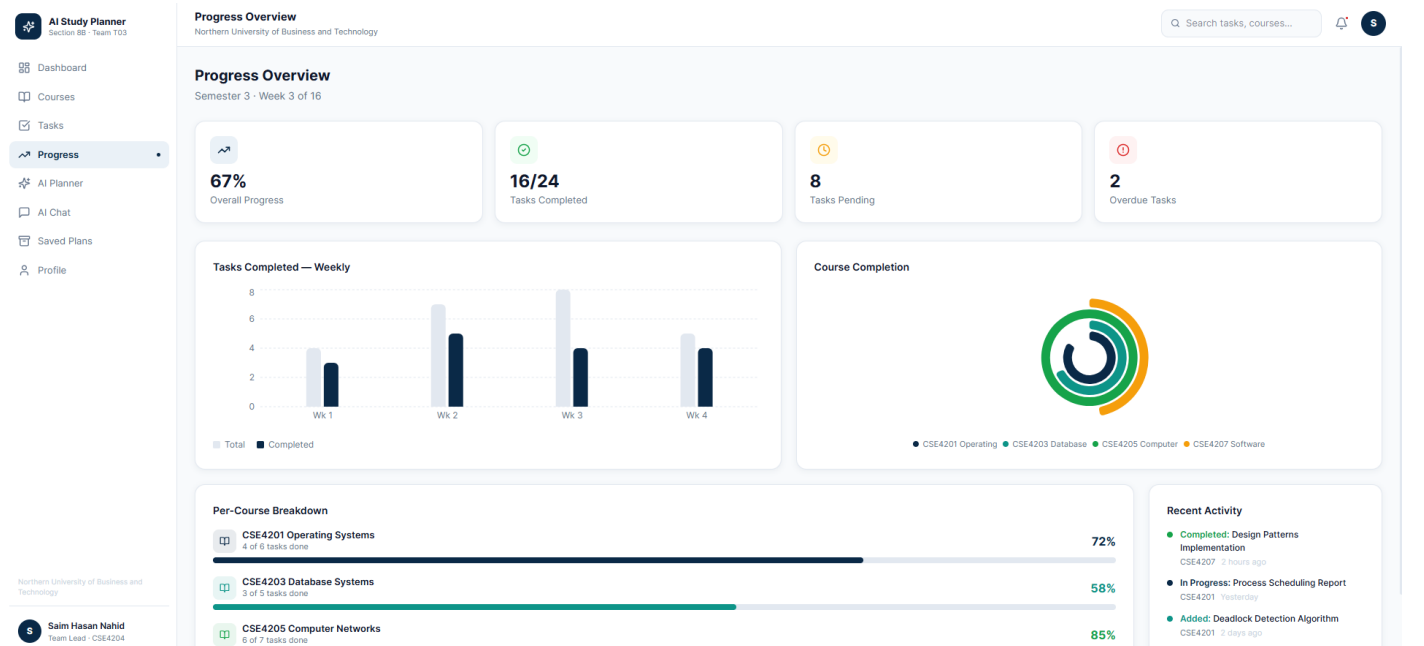


Figure 11. Weekly completion, course-wise progress, recent activity, pending, and overdue tasks.

UI DESIGN

AI Study Planner

The screenshot shows the 'AI Study Planner' interface. On the left is a sidebar with navigation links: Dashboard, Courses, Tasks, Progress, AI Planner (selected), AI Chat, Saved Plans, and Profile. The main header includes the user's name 'Salm Hasan Nahid' and a search bar. The main content area has tabs for 'Study Planner', 'Task Breakdown', and 'Priority Suggestion'. The 'Study Planner' tab is active, showing a 'Plan Configuration' section with four dropdown menus: 'Planning Duration' (7 Days), 'Focus Courses' (All (4 Courses)), 'Daily Study Hours' (4-6 hours), and 'Prioritise Tasks' (High priority first). A 'Generate Study Plan' button is at the bottom of the configuration section.

Figure 12. Study plan configuration using duration, focus courses, daily hours, and task priority.

AI Task Breakdown

The screenshot shows the 'AI Task Breakdown' interface. The sidebar is identical to the previous screen. The main header has a search bar and a user profile. The main content area has tabs for 'Study Planner', 'Task Breakdown' (selected), and 'Priority Suggestion'. The 'Task Breakdown' tab is active, showing a 'Select Task' section with three task cards: 'Process Scheduling Report' (CSE4201 - Due 20 Jun), 'ER Diagram Assignment' (CSE4203 - Due 25 Jun), and 'Network Protocol Lab' (CSE4205 - Due 22 Jun). A 'Break Down Task' button is at the bottom of the task selection section. The right side of the main content area is a large empty space with a faint AI icon and the text 'Select a task and click "Break Down Task" to get AI-generated steps'.

Figure 13. Task selection interface for generating smaller, actionable AI-supported steps.

UI DESIGN

AI Priority Suggestion

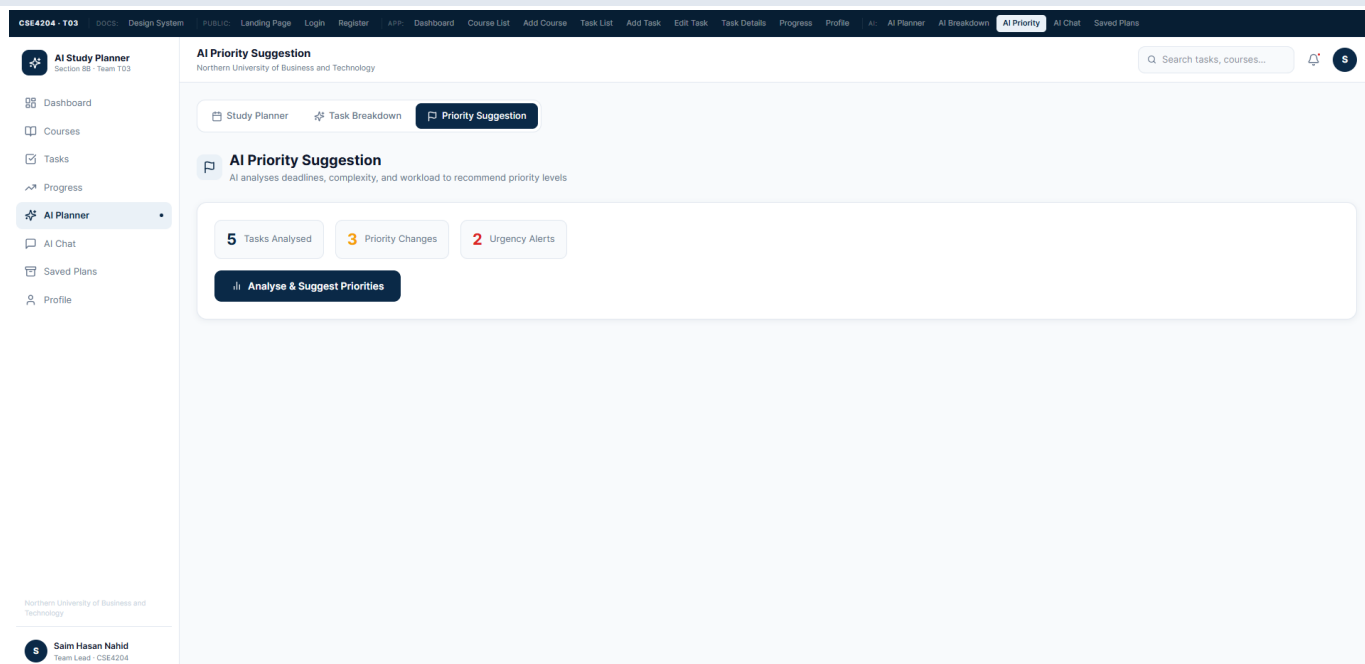


Figure 14. AI analysis of deadlines, workload, and complexity to recommend task priorities.

AI Study Assistant

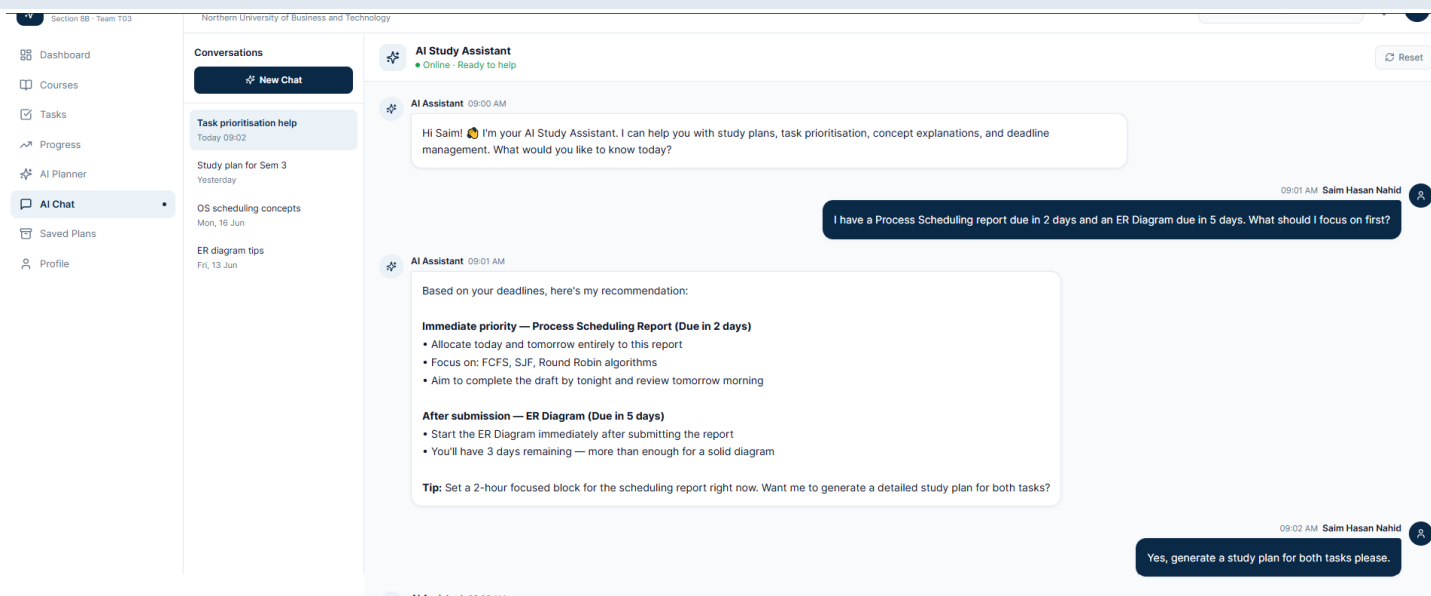


Figure 15. Conversational assistant for study plans, concepts, deadlines, and task prioritisation.

UI DESIGN

Saved AI Plans

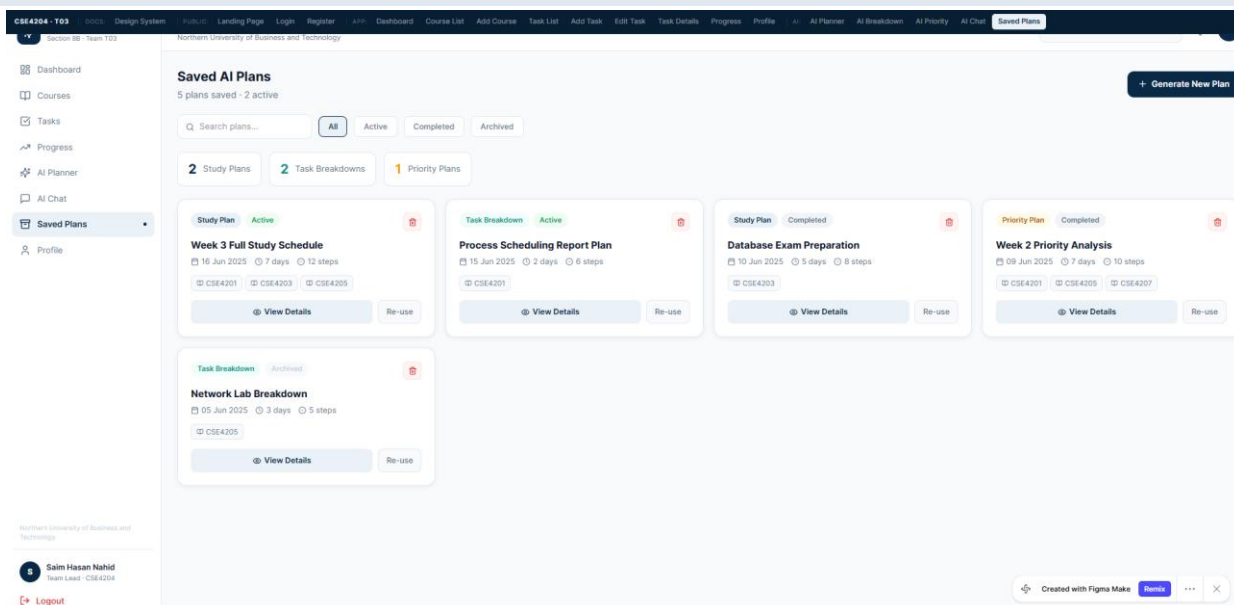


Figure 16. Searchable saved plans with active, completed, archived, detail, reuse, and delete actions.

User Profile

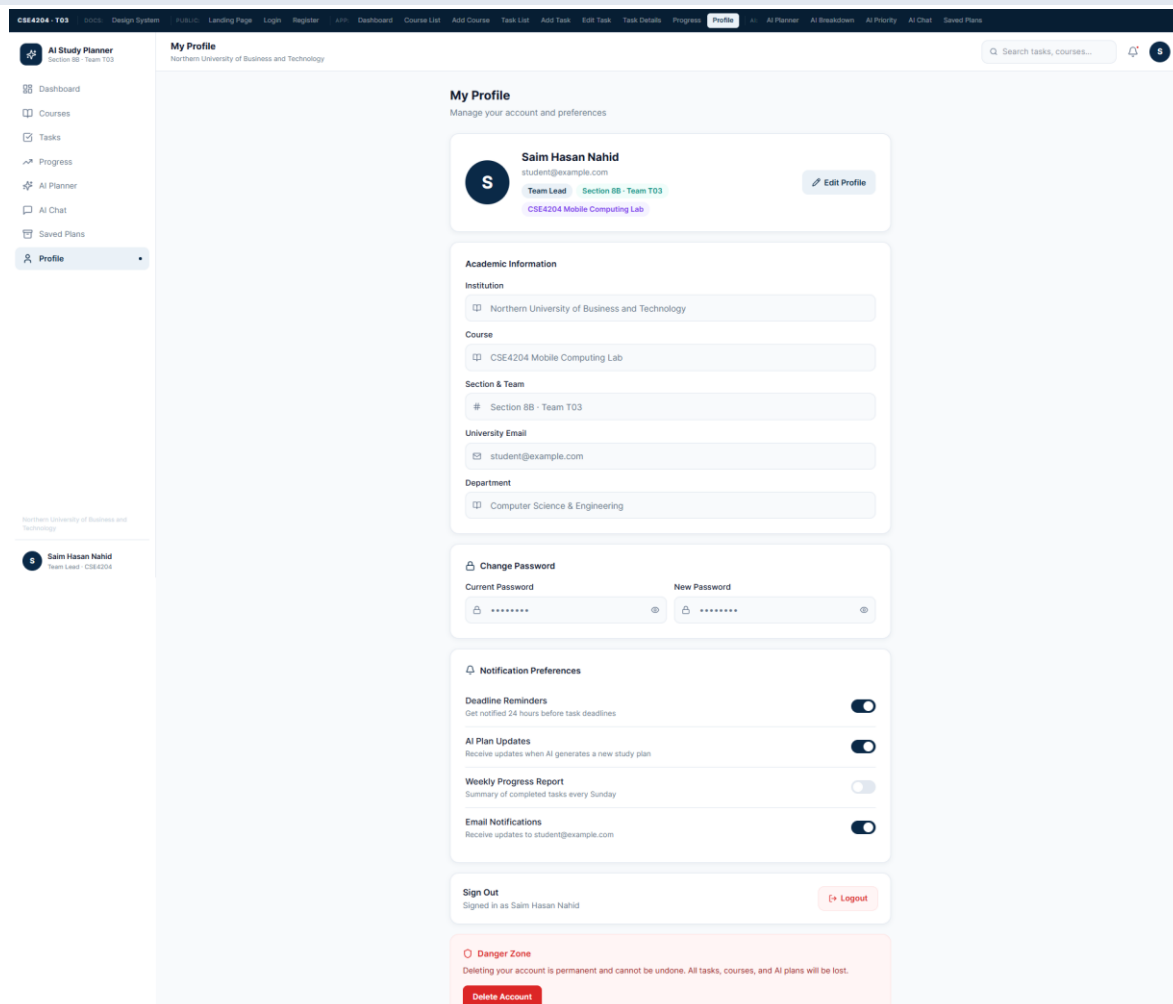


Figure 17. Student profile, academic information, password, notifications, logout, and account controls.

USER EXPERIENCE

User Flow Design

The following editable flow shows how a student navigates through the proposed system.

Step	Screen / Stage	Main User Action
1	Landing Page	Review the product and choose Get Started or Login.
2	Registration / Login	Create an account or sign in with university credentials.
3	Dashboard	Review courses, tasks, deadlines, progress, and quick actions.
4	Feature Access	Open Courses, Tasks, Progress, AI Planner, AI Chat, or Saved Plans.
5	AI Interaction	Provide task, course, schedule, or study information to the AI feature.
6	Results Display	Review generated plans, priorities, breakdowns, or chat responses.
7	Profile / Logout	Manage account preferences and securely end the session.

Feature Access Paths

Dashboard option	Typical next step
Courses	View courses -> add or edit a course -> open course tasks
Tasks	Search or filter tasks -> add/edit task -> mark complete
Progress	Review completion percentage, deadlines, and course-wise progress
AI Planner	Select data -> generate plan -> save or regenerate result
AI Chat	Ask questions -> receive recommendations -> continue conversation
Saved Plans	Open, reuse, archive, or delete previously generated plans

Flow Explanation

- New users register and existing users log in before accessing the application.
- The dashboard provides direct access to courses, tasks, progress, and AI features.
- AI-generated outputs can be reviewed, saved, reused, or regenerated.
- Users can manage account preferences and end the session from the profile area.

DEVELOPMENT PLANNING

Development Roadmap

Week	Phase	Planned Work
Week 06	Backend Development	Django project, DRF setup, authentication, course/task APIs, and PostgreSQL foundation.
Week 07	Frontend Development	React and Tailwind setup, screens, routing, forms, responsive components, and API connection.
Week 08	AI Integration	Gemini configuration, study plan, priority suggestion, task breakdown, chat, and saved plans.
Week 09	Feature Completion	Complete remaining modules, integration fixes, validation, state handling, and polish.
Week 10	Testing and Debugging	API tests, UI tests, responsive checks, security checks, error handling, and bug fixing.
Week 11	Deployment and Finalisation	Render backend, Netlify frontend, hosted PostgreSQL, README, demo, and submission.

TEAM RESPONSIBILITIES

Team Task Distribution

Member	Primary Responsibility	Planned Work
Saim Hasan Nahid	Team Lead and Integration	Repository management, project foundation, module integration, documentation, testing, deployment, and final review.
Aisha Siddika Oishee	Backend Development	Django/DRF APIs, authentication, database models, PostgreSQL verification, backend tests, and deployment support.
Eity	Frontend Development	Figma UI review, React pages, reusable components, responsiveness, forms, and frontend API integration.
Md Khaled Hasan Rihad	AI Integration	Gemini prompts, AI endpoints, study-plan logic, priority suggestions, task breakdown, chat testing, and response quality.

Collaboration and Quality Rule

- Each member validates and tests the assigned module before contribution.
- The Team Lead integrates contributions, resolves conflicts, performs end-to-end testing, and deploys stable versions.
- Documentation and GitHub progress are maintained throughout development.

TECHNOLOGY SETUP

Final Technology Stack and Setup Plan

Layer	Technology	Purpose
UI/UX	Figma Make	High-fidelity design, responsive planning, and prototype reference
Frontend	React.js + Tailwind CSS	Component-based responsive user interface
Backend	Django + Django REST Framework	Business logic and REST APIs
Authenticat tion	JWT / SimpleJWT	Token-based login and protected API access
Database	PostgreSQL	Persistent user, course, task, progress, and AI-plan data
AI Service	Gemini API	Study plans, priority suggestions, task breakdown, and chat
Testing	Postman + Browser Testing	API verification and UI behaviour testing
Deployme nt	Render + Netlify	Backend API and frontend web application hosting
Version Control	Git + GitHub	Collaboration, history, review, and submission

Planned Deployment Connection

